PYTHON VIRTUAL ENVIRONMENT GUIDE

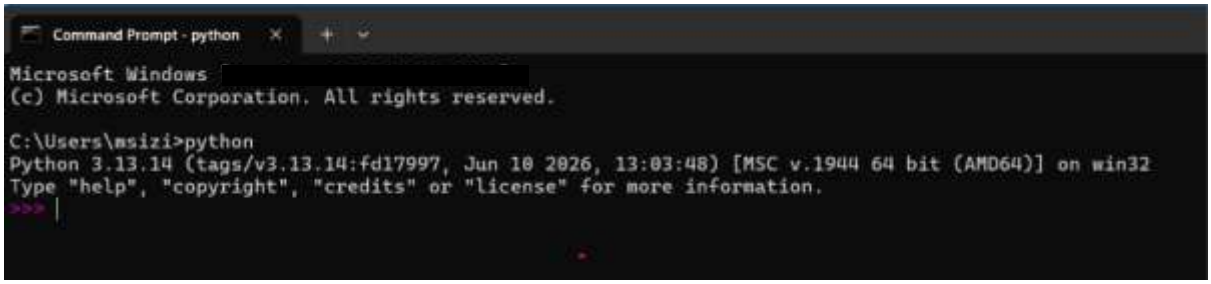
NB: This guide uses Windows Command Prompt (CMD). Commands for PowerShell, Linux, and macOS may differ.

What is a Python virtual environment?

It is an isolated, self-contained directory tree that contains its own Python interpreter, software libraries, and binaries required for a specific project. By creating this isolated container, you ensure that dependencies installed for one project will not interfere with system-wide packages or dependencies of other projects. But it uses an existing Python installation as its base.

Prerequisites and checking your Python installation

1. Open your Command Prompt to check if Python is installed.



If not, install the latest Python version on your machine. Check https://www.python.org/downloads/. After installing Python, you are free to move to the next step

Creating a project directory

1. Create a folder using CMD. E.g. Game or name it your own way.

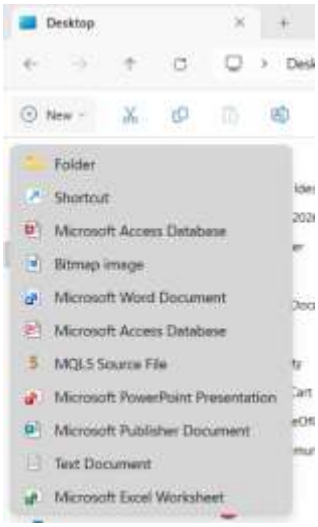
```
C:\Users\msizi>cd Desktop

C:\Users\msizi\Desktop>mkdir Game

C:\Users\msizi\Desktop>cd Game

C:\Users\msizi\Desktop\Game>
```

2. You can create it in your file explorer.



Creating a virtual environment

1. Since you are already in the game folder, you can run the command below. “myvenv” is the name of the virtual environment.

```
C:\Users\msizi\Desktop\Game>python -m venv myvenv

C:\Users\msizi\Desktop\Game>
```

2. Check if it was created; you should see the folder “myvenv”.

```
C:\Users\msizi\Desktop\Game>dir
 Volume in drive C has no label.
 Volume Serial Number is E4DC-D5FC

 Directory of C:\Users\msizi\Desktop\Game

2026/10/22  18:17    <DIR>          .
2026/10/22  18:06    <DIR>          ..
2026/10/22  18:17    <DIR>          myvenv
               0 File(s)              0 bytes
               3 Dir(s)  125 168 984 064 bytes free
```

Activating and deactivating the virtual environment

1. Activate your virtual environment

```
C:\Users\msizi\Desktop\Game>dir
 Volume in drive C has no label.
 Volume Serial Number is E4DC-D5FC

 Directory of C:\Users\msizi\Desktop\Game

2026/10/22  18:17    <DIR>          .
2026/10/22  18:06    <DIR>          ..
2026/10/22  18:17    <DIR>          myvenv
               0 File(s)              0 bytes
               3 Dir(s)  124 463 747 072 bytes free

C:\Users\msizi\Desktop\Game>myvenv\Scripts\activate.bat

(myvenv) C:\Users\msizi\Desktop\Game>
```

   If successful, your command prompt should look something like this:
   (myvenv) C:\Users\msizi\Desktop\Game>

2. Deactivate your virtual environment

```
C:\Users\msizi\Desktop\Game>myvenv\Scripts\activate.bat

(myvenv) C:\Users\msizi\Desktop\Game>deactivate
C:\Users\msizi\Desktop\Game>
```

Installing and managing packages with pip

1. Ensure that your virtual environment is active

```
C:\Users\msizi\Desktop\Game>myvenv\Scripts\activate.bat

(myvenv) C:\Users\msizi\Desktop\Game>pip install pygame
Collecting pygame
  Downloading pygame-2.6.1-cp313-cp313-win_amd64.whl.metadata (13 kB)
Downloading pygame-2.6.1-cp313-cp313-win_amd64.whl (10.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.6/10.6 MB 9.9 MB/s  0:00:01
Installing collected packages: pygame
Successfully installed pygame-2.6.1

[notice] A new release of pip is available: 26.1.2 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

2. You can also install a specific version of a package

```
(myvenv) C:\Users\msizi\Desktop\Game>pip install requests==2.6.0
Collecting requests==2.6.0
  Downloading requests-2.6.0-py2.py3-none-any.whl.metadata (31 kB)
Downloading requests-2.6.0-py2.py3-none-any.whl (469 kB)
Installing collected packages: requests
Successfully installed requests-2.6.0

[notice] A new release of pip is available: 26.1.2 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip

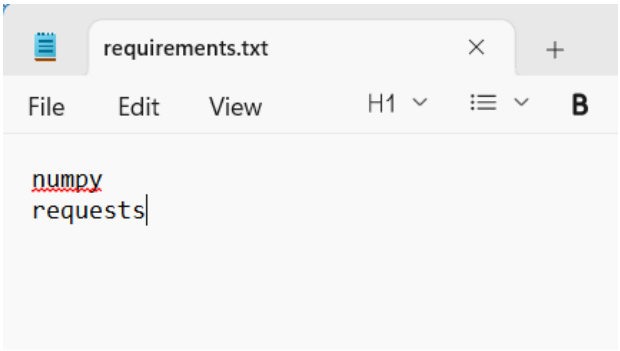
(myvenv) C:\Users\msizi\Desktop\Game>
```

Saving and reinstalling dependencies with requirements.txt

1. Inside the Game folder, create a file named requirements.txt and add the packages below



requirements.txt helps install multiple packages that you are going to use or that are in the current project you cloned from GitHub.



2. Pip freeze

pip freeze lists the packages installed in your current Python environment, including their versions.

```
(myvenv) C:\Users\msizi\Desktop\Game>pip freeze
numpy==2.5.4
pygame==2.6.1
requests==2.6.0

(myvenv) C:\Users\msizi\Desktop\Game>
```

Save the list to requirements.txt

- python -m pip freeze lists the installed packages and their versions.
- > redirects the output into a file.
- requirements.txt is the file where that list is saved.
  Important: > creates the file if it doesn't exist but overwrites it if it already exists.

```
(myvenv) C:\Users\msizi\Desktop\Game>pip freeze > requirements.txt
```

3. Why is this useful?
   Imagine you develop a Python project on your computer and want a friend to run it on theirs.

Troubleshooting common errors

Spelling mistakes exist, so keep calm.

```
(myvenv) C:\Users\msizi\Desktop\Game>pip install -r requirements.txt

[notice] A new release of pip is available: 26.1.2 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip
ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements.txt'

(myvenv) C:\Users\msizi\Desktop\Game>dir
 Volume in drive C has no label.
 Volume Serial Number is E4DC-D5FC

 Directory of C:\Users\msizi\Desktop\Game

2026/10/22  19:09    <DIR>          .
2026/10/22  18:06    <DIR>          ..
2026/10/22  18:17    <DIR>          myvenv
2026/10/22  19:13                45 requirements.txt.txt
               1 File(s)             45 bytes
               3 Dir(s)  123 404 312 576 bytes free
```

- **Activation fails: check that myvenv\Scripts\activate.bat exists and that CMD is in the project directory.**
- **python is not recognised: check the Python installation and PATH configuration; on Windows, try py --version.**
- **pip cannot install a package: check the error message, Python version, and package compatibility.**
- **requirements.txt is not found: check that CMD is in the directory containing the file.**
- **The wrong Python is being used: run python -c "import sys; print(sys.executable)".**

References

https://docs.python.org/3/tutorial/venv.html

https://www.python.org/downloads/

https://docs.python.org/3/library/venv.html

https://pip.pypa.io/en/stable/user_guide/